

Module 12: Configuring Power Management

◇ What is Power Management in SCCM?

- A feature that helps IT admins **control computer power usage**.
- Saves electricity  and reduces cost.
- Ensures all devices follow **standardized power-saving rules**.

◆ Key Features

1. **Power Plans** → Define rules for sleep, hibernate, screen off, etc.
2. **Schedule Power Settings** → Different for work hours vs after-hours.
3. **Reports** → Measure power usage and savings.

◆ Example Scenarios

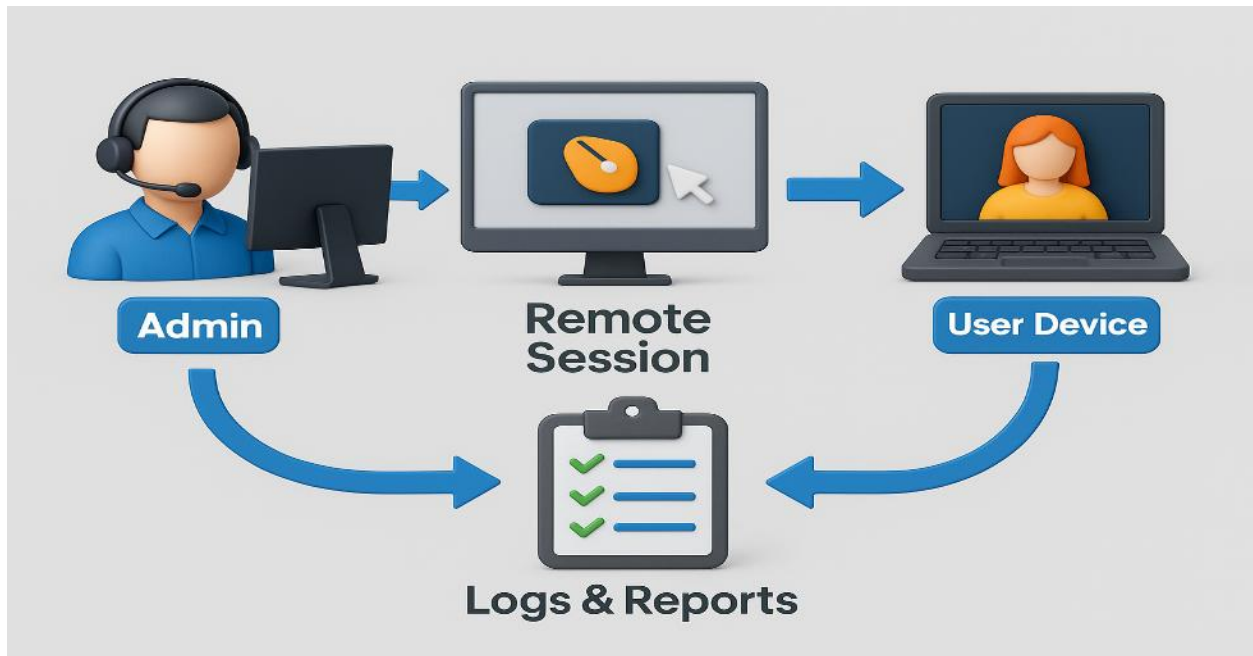
- Office desktops → Sleep at **8:00 PM** if idle.
- Laptops → Screen dims after **10 minutes** inactivity.

◆ Steps to Configure in SCCM

1. Open **SCCM Console** → **Administration** → **Client Settings** → **Default Client Settings**.
2. Enable **Power Management**.
3. Create **Custom Power Plans** (e.g., “Work Hours” & “After Hours”).
4. Deploy these plans to **device collections** (like All Laptops, All Desktops).

☒ **Result:**

- All devices follow the same **power-saving rules**.
- IT admins can **track savings** in reports.



2. Configuring Remote Control in SCCM

◆ What is Remote Control?

- A **built-in SCCM feature** that allows IT admins to **view or control** end-user devices remotely.
- Mainly used for **support, troubleshooting, and training**.

◆ Key Features

1. **Remote Session** → IT can take control or just view the user's screen.
2. **Permission Settings** → Can be configured to ask user's approval before starting.
3. **Logging & Auditing** → Every remote session is recorded in logs for security.
4. **Role-Based Access** → Only authorized IT staff can use remote control.

◆ Steps to Configure Remote Control in SCCM

1. Open **SCCM Console** → **Administration** → **Client Settings** → **Default Client Settings**.
2. Go to **Remote Tools**.
3. Enable **Remote Control**.
4. Configure options:
 - Ask for user's permission (Yes/No).
 - Enable **logging** and **auditing**.
 - Restrict which admins/groups can use it.
5. Deploy the settings to device collections.

◆ Example Scenario

- An employee reports a software crash.
- IT admin opens SCCM console → Connects remotely to user's PC.
- Fixes issue instantly without visiting the location.

☒ Result:

- Quick, secure, and logged **remote troubleshooting**.
- Saves travel time and increases IT support efficiency.



Bilkul Ankit 👍 main aapko ek step-by-step lab exercise (practical guide) deta hoon jisme aap Configuring Remote Control in SCCM ko practice kar sakte ho.

🔧 Lab Exercise: Configuring Remote Control in SCCM

Step 1: Open the SCCM Console

1. Log in to your **SCCM Primary Site Server**.
2. Open the **Configuration Manager Console**.

Step 2: Go to Client Settings

1. Navigate **Administration** → **Client Settings** → **Default Client Settings** to:
2. Right-click **Default Client Settings** → Select **Properties**.

Step 3: Configure Remote Tools

1. In the settings window, go to the **Remote Tools** tab.
2. Configure the following:

- ☒ Enable **Remote Control** → **Yes**
- ☒ Allow **View Only** → Optional (for monitoring only)
- ☒ Prompt user for approval → **Yes/No** (based on policy)
- ☒ Enable **Auditing** → **Yes** (so every session is logged)
- ☒ Show notification icon to user → **Yes**

Step 4: Security Permissions

1. Go to **Administration** → **Security** → **Security Roles**.
2. Make sure **Helpdesk/IT Support** role has **Remote Control** permission.
3. Assign roles to the correct admin users.

Step 5: Deploy Client Settings

1. Right-click the configured **Client Settings**.
2. Select **Deploy**.
3. Choose the **Device Collection** (e.g., "All Workstations").

Step 6: Test Remote Control

1. From **Assets and Compliance** → **Devices**, right-click a client computer.
2. Select **Start** → **Remote Control**.
3. If configured:
 - User sees a **prompt** → clicks **Yes**.
 - Admin gains remote access.

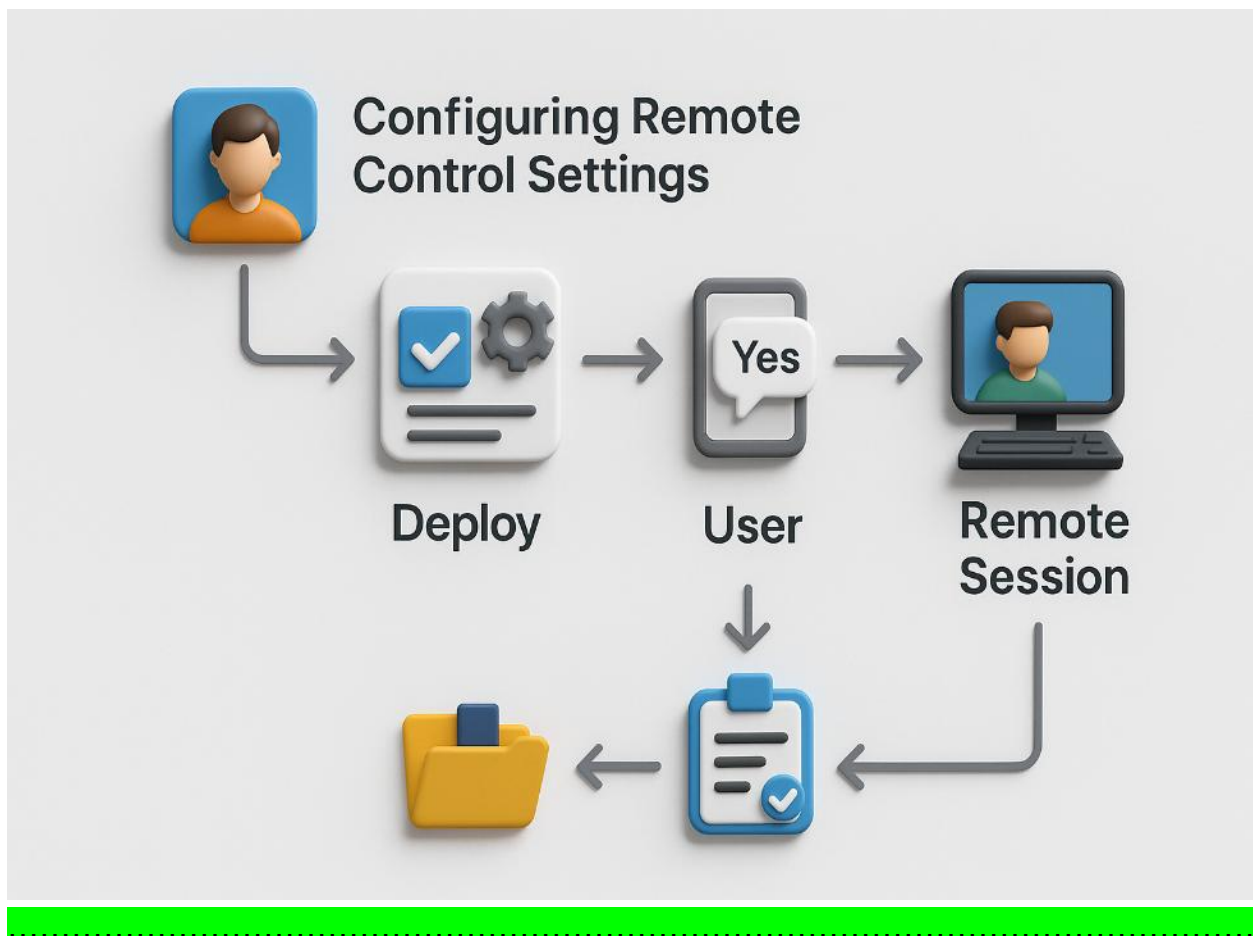
Step 7: Verify Logs

Check logs to confirm remote session:

- On **Client PC** → **CmRcService.log**
- On **SCCM Server** → Session logs under Remote Control reports

✅ End Result:

- Remote Control is enabled.
- IT admins can securely support users.
- All sessions are tracked in logs & reports.



Module 13: Overview of Configuration Manager Site Maintenance

1. Performing Backup And Recovery Of A System Center Configuration Manager (SCCM) Site

◆ Why Site Maintenance is Important?

- Keeps the SCCM environment healthy, secure, and recoverable.
- Ensures that if something goes wrong (like server crash, database corruption, or hardware failure), you can restore SCCM and continue operations.

1. Performing Backup of an SCCM Site

◇ Backup Types in SCCM:

1. Site Database Backup (SQL) – Most important (stores all data about clients, deployments, settings).
2. SCCM Site Server Backup – Includes configuration, packages, roles.
3. Content Library Backup – Holds applications, updates, OS images, etc.

◇ Methods to Backup

1. Built-in SCCM Backup Maintenance Task

- Configured in SCCM console under Administration → Site Configuration → Sites → Site Maintenance.
- Creates a scheduled backup of SCCM site + SQL DB.

2. SQL Server Backup

- Directly backup SCCM DB from SQL Server Management Studio.

3. Manual File System Backup

- Backup SCCM installation folder, **Inboxes**, **Logs**, and **Content Library**.

◇ Backup Location

- Store backups on a separate drive or remote server (not same SCCM server).
- For disaster recovery, keep an offsite/cloud backup copy.

2. Performing Recovery of an SCCM Site

If SCCM crashes or database is lost, recovery methods:

- | 1. Site | Recovery | Wizard | (built-in) |
|---------------------------------|----------|--|------------|
| ○ Run from SCCM setup | → | Choose Recover a Site. | |
| ○ Options: | | | |
| ■ Recover using a backup set | → | Restore site DB & settings. | |
| ■ Reinstall site server | → | Rebuild roles and reattach DB. | |
| 2. SQL | | | Restore |
| ○ Restore SCCM database in SQL, | | then re-sync with SCCM. | |
| 3. Manual | | | Recovery |
| ○ Reinstall SCCM, | | reconfigure roles, then re-import backups. | |

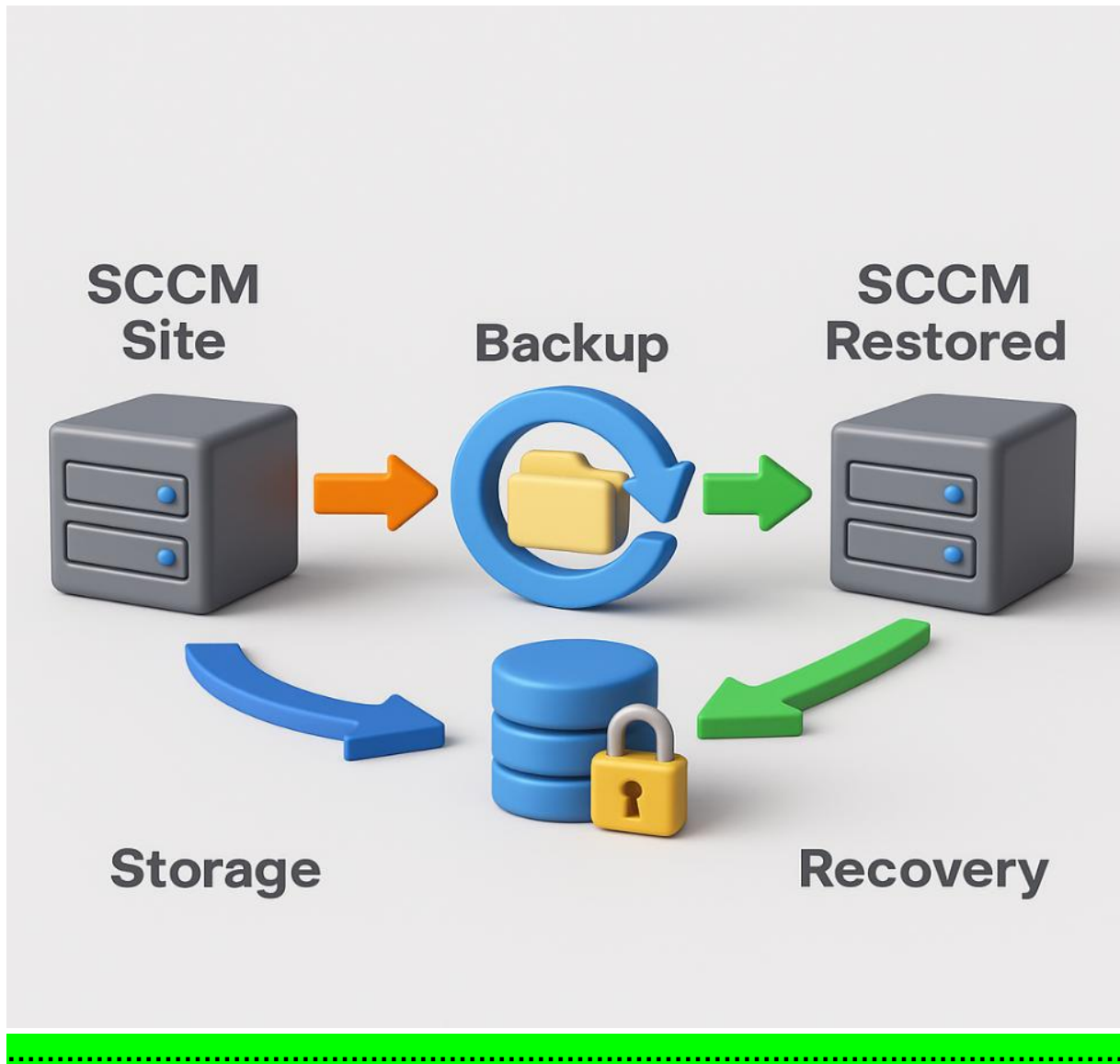
☑ Example Scenario

- A company's SCCM server crashes due to hardware failure.
- IT runs Site Recovery Wizard → Selects latest backup from remote server → SCCM reinstalled with same settings, SQL DB restored → Services back online in few hours.

☑ Summary

- Backup = Regularly save site DB, content library, and site server files.

- Recovery = Use Site Recovery Wizard or SQL restore to bring SCCM back.
- Always test recovery plan to ensure business continuity.



2. Monitoring System Center Configuration Manager (SCCM) Site Systems

- ◆ What is Site System Monitoring?

- In SCCM, a site system is any server role (like Management Point, Distribution Point, Software Update Point, Reporting Services Point).
- Monitoring site systems means checking their health, availability, and performance so that SCCM runs smoothly.

◆ Key Components to Monitor

1. **Site Server** → The main SCCM server hosting the database and roles.
2. **Management Point (MP)** → Communicates with clients; must be healthy.
3. **Distribution Point (DP)** → Stores and distributes content (apps, updates, OS images).
4. **Software Update Point (SUP)** → Syncs patches from Microsoft.
5. **Reporting Services Point (RSP)** → Generates reports.

◇ Monitoring Tools in SCCM

1. **SCCM Console** → Monitoring Workspace
 - Status of site roles (healthy/unhealthy).
 - Site system status nodes.
2. **Alerts & Notifications**
 - Critical site component failure alerts.
 - Alerts for disk space, sync failure, etc.
3. **Logs**
 - Each role has its own log file (e.g., `mpcontrol.log` for Management Point).

4. Reports

- Built-in reports for role availability and performance.

◆ Example Scenarios

- If the Distribution Point (DP) runs out of disk space → software deployments fail.
- If the Management Point (MP) is down → clients cannot report status.
- Monitoring catches these issues early so IT can fix them.

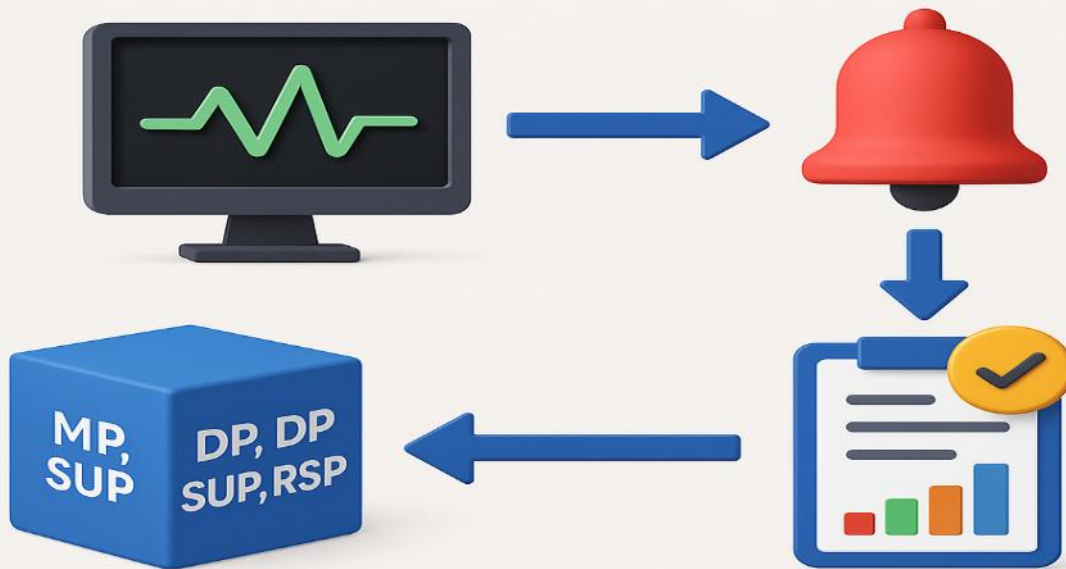
◆ Steps to Monitor Site Systems

1. Open SCCM Console → Monitoring → System Status.
2. Check Site Status (overall health).
3. Check Component Status (individual roles).
4. Use Reports for historical health data.
5. Review critical logs for troubleshooting.

☒ Result:

- Ensures SCCM infrastructure is healthy.
- Prevents failures in app deployment, updates, OS deployment.
- Helps IT quickly detect and fix issues.

Monitoring System Center Configuration Manager (SCCM) Site Systems



Module 14: Addition of a New Secondary Site

1. Install A New Secondary Site

◆ What is a Secondary Site in SCCM?

- A Secondary Site is used in branch offices or remote locations where network bandwidth to the Primary Site is slow or limited.
- It helps manage clients in that location without needing a full Primary Site.

👉 Example:

- Headquarters in Delhi = Primary Site

- **Branch office in Lucknow = Secondary Site** (to manage local clients efficiently).

◆ **Why Use a Secondary Site?**

1. To reduce WAN traffic between branch office and primary site.
2. To keep a local Distribution Point (DP) for apps/updates.
3. To manage clients without creating another full Primary Site.

◆ **Requirements for Installing a Secondary Site**

1. You must already have a Primary Site.
2. Windows Server installed on the branch office server.
3. SQL Server Express (installed automatically if not present).
4. Network connectivity with Primary Site.
5. Proper firewall ports opened.

◆ **Steps to Install a Secondary Site**

1. Open SCCM Console → Administration → Site Configuration → Sites.
2. Right-click Sites → Select Create Secondary Site.
3. Provide:
 - Server name (branch office server).
 - Site Code & Site Name (e.g., LKO = Lucknow).
4. **Configure SQL Server** (SCCM can install SQL Express automatically).
5. **Configure Content Library** (storage path for apps, updates, OSD images).

6. Define Communication Settings (with Primary Site).

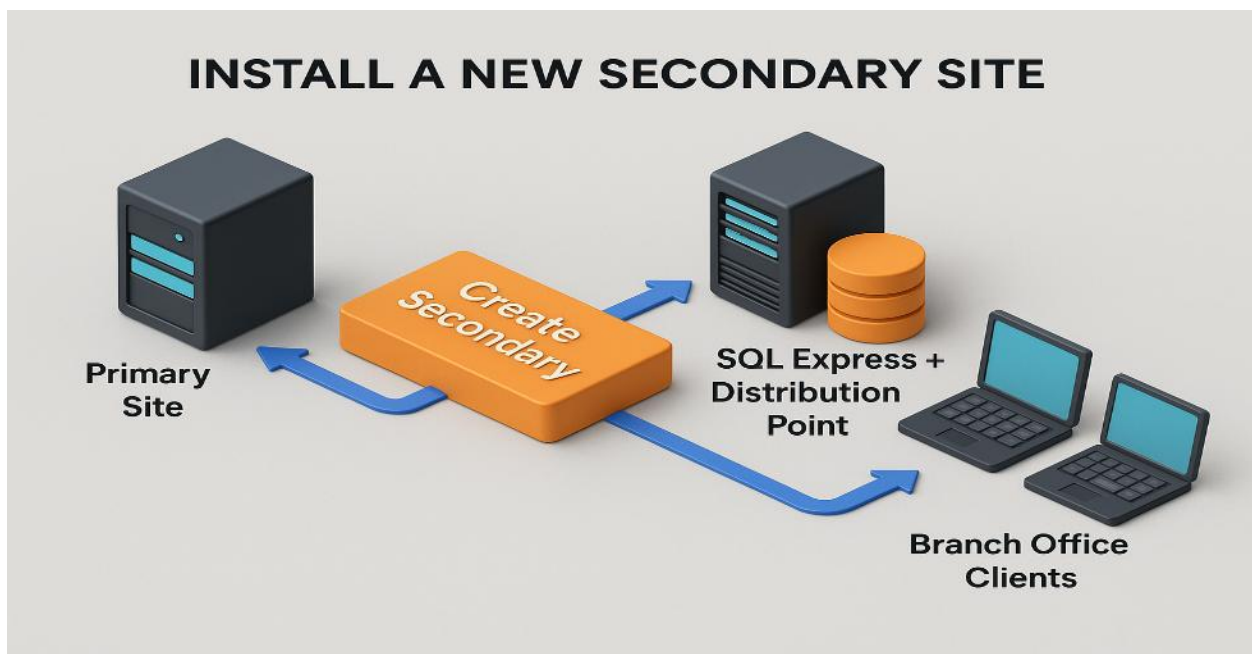
7. Finish installation → Monitor progress in **ConfigMgrSetup.log**.

◆ Example Scenario

- A company has 200 PCs in a branch office connected via a slow WAN link.
- Instead of downloading updates from Delhi HQ DP, clients download locally from Lucknow Secondary Site DP.
- This saves bandwidth and improves speed.

✓ Summary

- A Secondary Site is dependent on a Primary Site.
- Provides local content distribution and better management for branch clients.
- Installed via SCCM console → Create Secondary Site wizard.



2. Assign Devices to Point to the Secondary Site Using Boundary and Boundary Group

♦ Why Boundaries & Boundary Groups?

- **Boundary** = Defines the network location of clients (like IP Subnet, AD Site, IP Range).
- **Boundary Group** = A logical collection of boundaries.
- SCCM uses boundary groups to **assign clients to sites and distribution points (DPs)**.

👉 In the case of a **Secondary Site**, we create boundaries for the **branch office network** and assign them to that site.

♦ Example

- **Delhi HQ (Primary Site):** 192.168.1.0/24
- **Lucknow Branch (Secondary Site):** 192.168.2.0/24

Here, devices in the **192.168.2.0/24** subnet should connect to the **Lucknow Secondary Site**.

♦ Steps to Assign Devices to Secondary Site

1. Open **SCCM Console** → **Administration** → **Hierarchy Configuration** → **Boundaries**.
2. Create a new **Boundary** for the branch office (e.g., IP Subnet = 192.168.2.0/24).
3. Go to **Boundary Groups** → Create new group (e.g., Lucknow_Branch).
4. Add the boundary (192.168.2.0/24) to this group.
5. In the **References** tab, specify:

- **Secondary Site Server** (Lucknow SCCM)
- **Distribution Point** (local DP in branch).

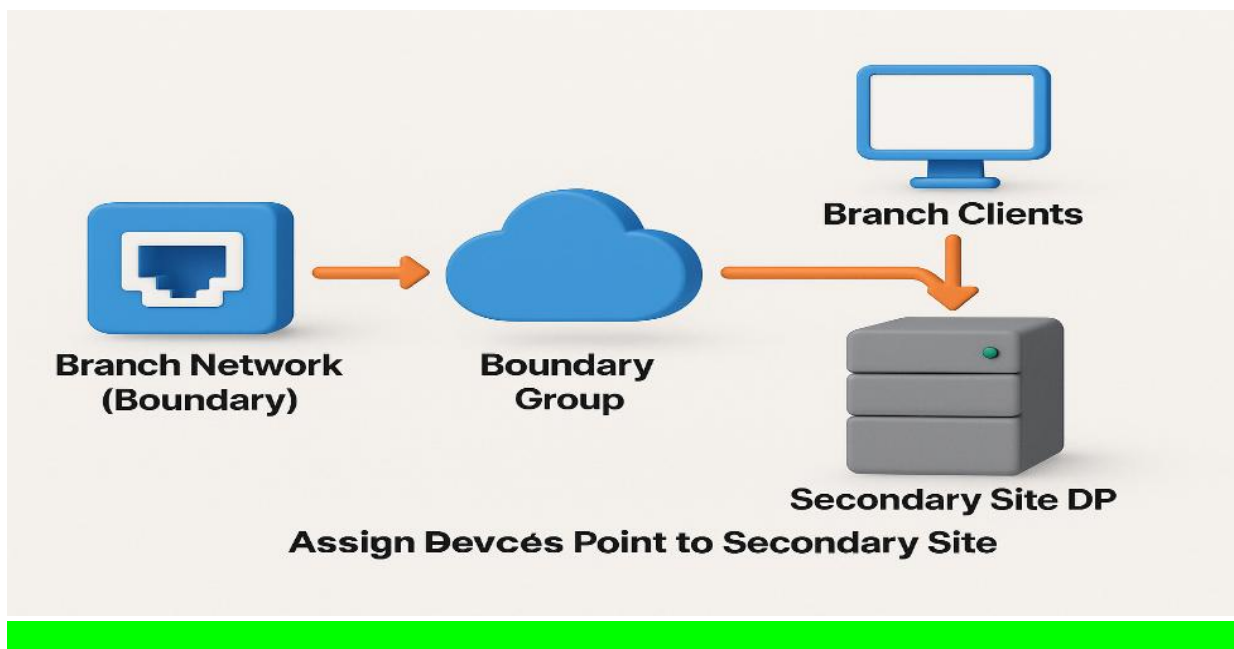
6. Deploy policies or applications → Devices in Lucknow subnet will automatically get content from the **Secondary Site DP**.

◇ **Benefits**

- Ensures branch office clients use **local Secondary Site DP** instead of WAN link.
- Reduces network load.
- Faster deployments and updates.

☑ **Summary:**

- **Boundary** = Network location (IP Subnet/Range).
- **Boundary Group** = Assigns devices to Secondary Site + DP.
- Clients automatically talk to the nearest **site server/DP**.



3. Deploy Applications to Devices Through Secondary Site DP

◆ How It Works

When you install a **Secondary Site**, it comes with a **local Distribution Point (DP)**.

- Primary Site → Sends application/package content to Secondary Site DP.
- Branch Clients → Download apps/updates **locally** from the Secondary Site DP instead of WAN.

👉 This reduces **bandwidth usage** and improves **speed** for branch users.

◇ Example

- **Primary Site (Delhi)** hosts **Office 365 app package**.
- Content is distributed to **Secondary Site DP (Lucknow)**.
- Devices in Lucknow (branch office) get Office 365 from **Lucknow DP** instead of pulling it from Delhi over WAN.

◆ Steps to Deploy Applications via Secondary Site DP

1. **Distribute Content to Secondary Site DP**
 - In SCCM Console → Go to **Software Library** → **Applications**.
 - Right-click application → Select **Distribute Content**.
 - Choose **Secondary Site DP** (Lucknow DP).
2. **Deploy Application to Branch Devices**
 - Create a deployment → Target **Device Collection** of Lucknow PCs.
 - Select **install** action (Required/Available).
3. **Client Downloads Locally**

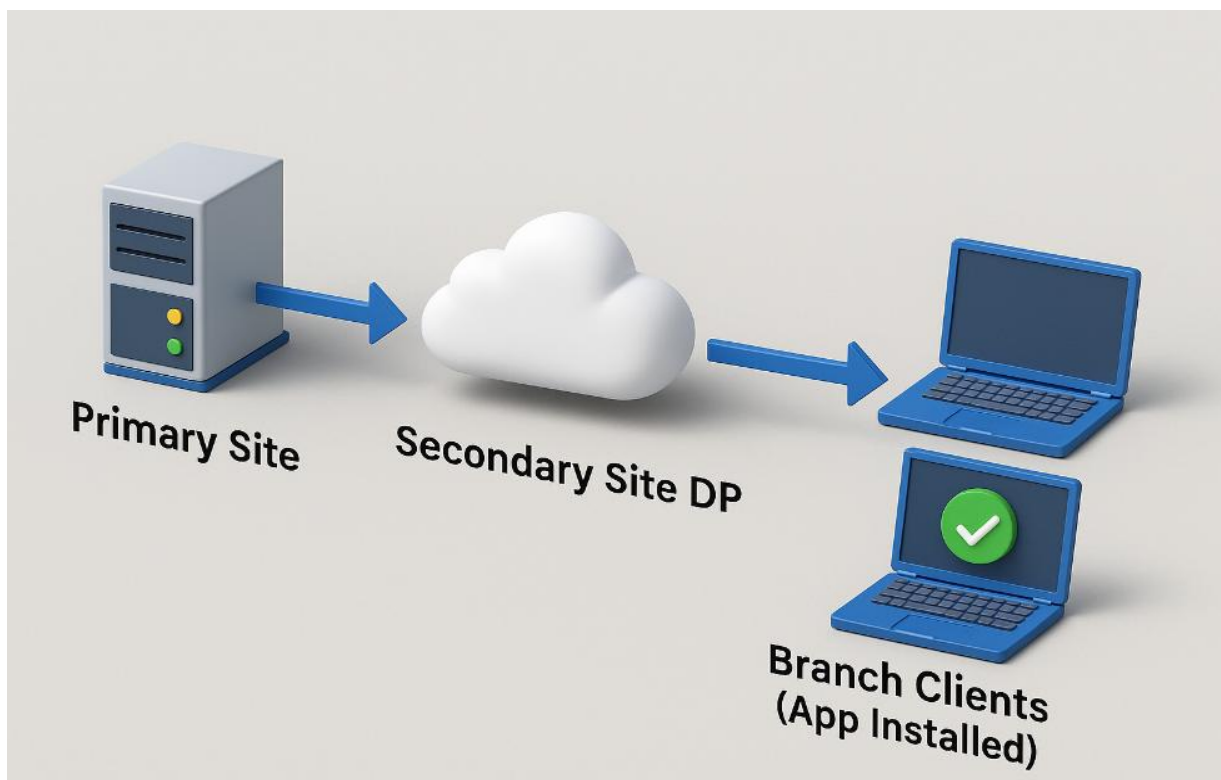
- Branch clients detect their **Boundary Group**.
- They connect to **Secondary Site DP** for faster download.

◇ **Benefits**

- Saves WAN bandwidth between Primary and Secondary Site.
- Faster app deployments at branch locations.
- Local distribution keeps operations **independent of HQ connectivity**.

☑ **Summary:**

- Apps first → sent from **Primary Site** → **Secondary Site DP**.
- Branch devices then → download from **local DP**.
- Improves **speed + efficiency** of deployments.



4. Viewing Log Files, Monitoring, and Troubleshooting (Secondary Site)

◆ Why Logs and Monitoring are Important?

- Secondary Sites depend on Primary Sites.
- If something goes wrong (like DP content not syncing, SQL Express failure, or boundary misconfiguration), SCCM generates logs and alerts.
- Monitoring + Logs = Helps IT quickly identify and fix issues.

◆ Key Log Files for Secondary Site

1. On Secondary Site Server:

- `ConfigMgrSetup.log` → Installation progress & errors
- `smsexec.log` → SCCM services running on site
- `sender.log` → Replication between Primary & Secondary Site
- `distmgr.log` → Content distribution to DP
- `hman.log` → Hierarchy manager (site communication)

2. On Client Machines (Branch Office):

- `LocationServices.log` → Finds nearest site/DP
- `AppDiscovery.log` → Application detection
- `CAS.log` → Content download status
- `AppEnforce.log` → Application install/uninstall

◆ Monitoring in SCCM Console

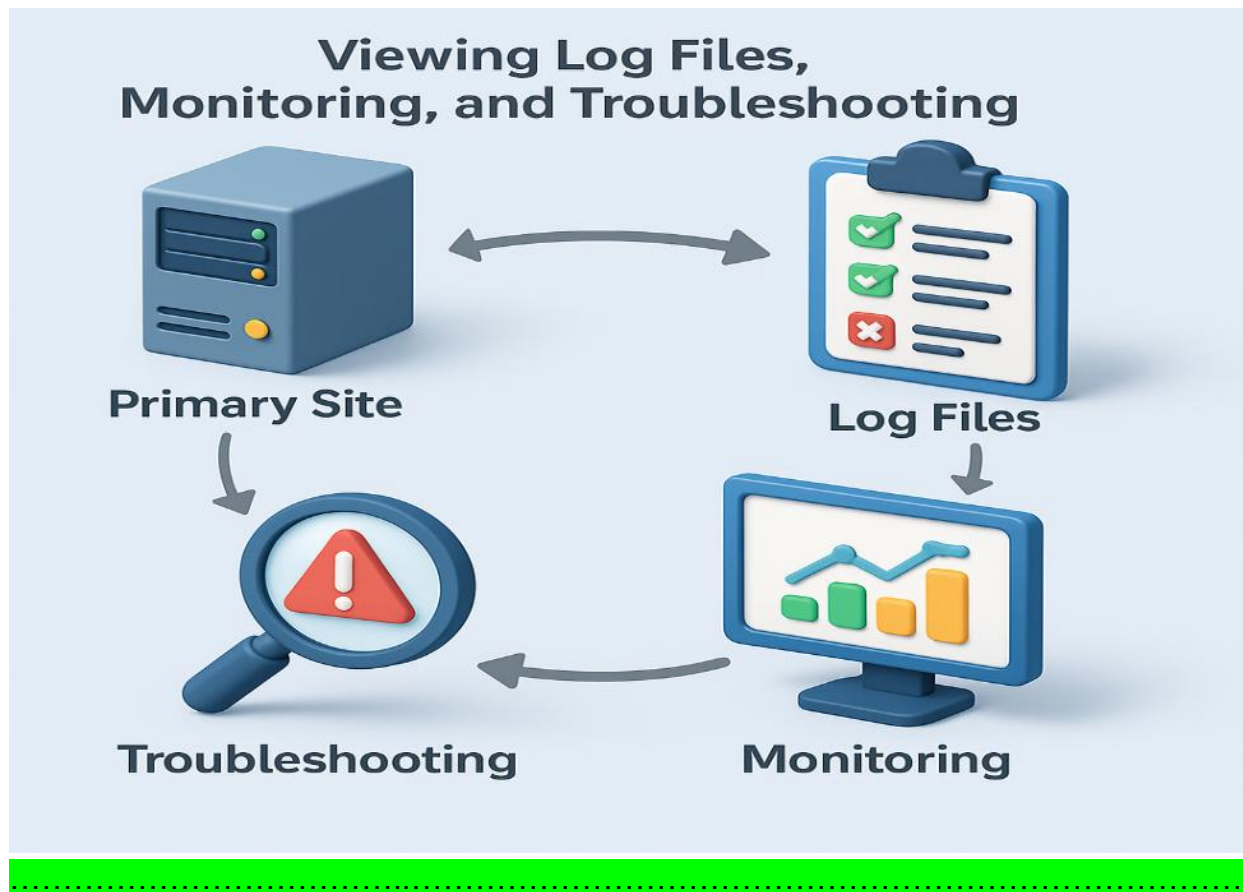
1. Open Monitoring Workspace.
2. Check Site Status → Health of Primary & Secondary Sites.
3. Check Component Status → Distribution Point, Management Point, SQL replication.
4. Use built-in Reports → For deployment success/failure, client assignment, boundary usage.

◇ Troubleshooting Common Issues

1. App not downloading in Branch:
 - Check if content is distributed to Secondary Site DP ([distmgr.log](#)).
 - Verify Boundary & Boundary Group mapping.
2. Replication issue between Primary & Secondary Site:
 - Check [sender.log](#) and [hman.log](#).
3. Client not installing app:
 - Check logs on client ([AppDiscovery.log](#), [AppEnforce.log](#)).

☑ Summary:

- Logs = Detailed troubleshooting (server & client side).
- Monitoring Workspace = Site & role health.
- Reports = Deployment & system compliance visibility.
- Helps admins detect and fix issues quickly in Secondary Sites.



Module 15: Miscellaneous Topics

1. Maintenance Window Configuration

What is a Maintenance Window?

- A **Maintenance Window** in SCCM is a **time period** during which specific actions (like software updates, app deployments, or OS upgrades) are allowed to run on client devices.
- Outside the window → deployments are **paused/delayed**.

👉 Example:

- Company doesn't want updates to install during office hours (9 AM – 6 PM).
- IT creates a **Maintenance Window: 7 PM – 11 PM**.

- Updates and reboots happen only at night.

◆ Why Use Maintenance Windows?

1. Avoid disruption during working hours.
2. Control **when reboots** happen after updates.
3. Ensure compliance without impacting productivity.

◆ Types of Deployments Affected

- **Software Updates**
- **Applications/Packages**
- **Task Sequences (OSD, in-place upgrades)**
- **Compliance settings**

◆ Steps to Configure Maintenance Window in SCCM

1. Open **SCCM Console** → **Assets and Compliance** → **Device Collections**.
2. Right-click the **Collection** (e.g., All Servers / HR Dept PCs).
3. Select **Properties** → **Maintenance Windows** tab.
4. Click **New Maintenance Window**.
5. Set:
 - Start Time & End Time
 - Recurrence (One time, Daily, Weekly, Monthly)
 - Apply to → Software Updates, Deployments, Task Sequences

6. Save and apply.

◆ **Example Scenario**

- **Servers** → Maintenance Window = Every Sunday 1 AM – 4 AM.
- **Workstations** → Maintenance Window = Weekdays 8 PM – 11 PM.

This ensures updates happen **after work hours**, reducing downtime.

☒ **Summary:**

- Maintenance Windows = Scheduled time for deployments.
- Prevents user disruption during business hours.
- Can be configured per **collection**.



2. Role-Based Administration in SCCM

◆ What is Role-Based Administration (RBA)?

- **RBA** controls **who can do what** in the SCCM Console.
- It ensures IT staff only have access to the features and objects they are responsible for.
- Helps improve **security** and prevents mistakes by limiting unnecessary access.

👉 Example:

- A **Helpdesk Engineer** can only remote control devices, not deploy software.
- A **Patch Admin** can manage Windows Updates but cannot modify collections.

◆ Key Elements of RBA

1. Security

Roles

- Define **what actions** a user can perform.
- Examples: Application Administrator, Compliance Settings Manager, Read-only Analyst.

2. Security

Scopes

- Define **which objects** a user can manage.
- Example: An admin can only manage collections in **HR Department**.

3. Collections

- Groups of devices/users that permissions apply to.
- Example: An admin can only deploy to the “**Marketing PCs**” collection.

◆ How It Works Together

- Security Role → Defines **permissions**.
- Security Scope → Defines **objects/resources**.
- Collection → Defines **target devices/users**.

👉 Combined = User gets **specific access only**.

◆ Steps to Configure Role-Based Administration

1. Open **SCCM Console** → **Administration** → **Security** → **Administrative Users**.
2. Add a new user/group from Active Directory.
3. Assign:
 - **Security Role** (what they can do).
 - **Security Scope** (which objects).
 - **Collection** (which devices/users).
4. Save and apply.

◆ Example Scenario

- **Helpdesk Team** → Assigned “Read-only Analyst” role + “All Workstations” collection.
- **Patch Admin** → Assigned “Software Update Manager” role + “Servers” collection.
- **Application Admin** → Assigned “Application Administrator” role + “HR Department PCs”.

☑ **Summary:**

- RBA = Secure, controlled access to SCCM.
- Roles = What actions → Scopes = Which objects → Collections = Which devices/users.
- Prevents misuse, improves compliance & security.



3. Accessing SCCM Console from a Different Computer

◆ What is the SCCM Console?

- The **SCCM (Configuration Manager) Console** is the main tool used by admins to manage applications, updates, OS deployments, compliance, and reporting.

- By default, the console is installed only on the **Primary Site Server**.
- But admins can also install it on their own **workstations/laptops** for remote management.

◆ **Why Install Console on Another Computer?**

1. IT admins don't need to log in to the SCCM server every time.
2. Multiple admins can use SCCM at the same time from their own PCs.
3. Improves security – avoids giving all admins direct access to the site server.

◆ **Prerequisites**

- Windows 10/11 (or Server) machine
- .NET Framework installed
- SQL Server Native Client installed
- Proper permissions in SCCM (Role-Based Administration)
- Network connectivity to the SCCM site server

◆ **Steps to Install SCCM Console on Another Computer**

Go to the **SCCM Server** → Open folder:

`<SCCM Installation Directory>\Tools\ConsoleSetup`

1. Example: `\\SCCMServer\SMS_<SiteCode>\tools\ConsoleSetup`
2. Run the **ConsoleSetup.exe** file.
3. Provide the **Site Server Name** (e.g., `SCCM-PRI.domain.com`).
4. Complete installation → Console shortcut appears on the workstation.

5. Launch the SCCM Console → Log in with **your AD account** (permissions defined via Role-Based Administration).

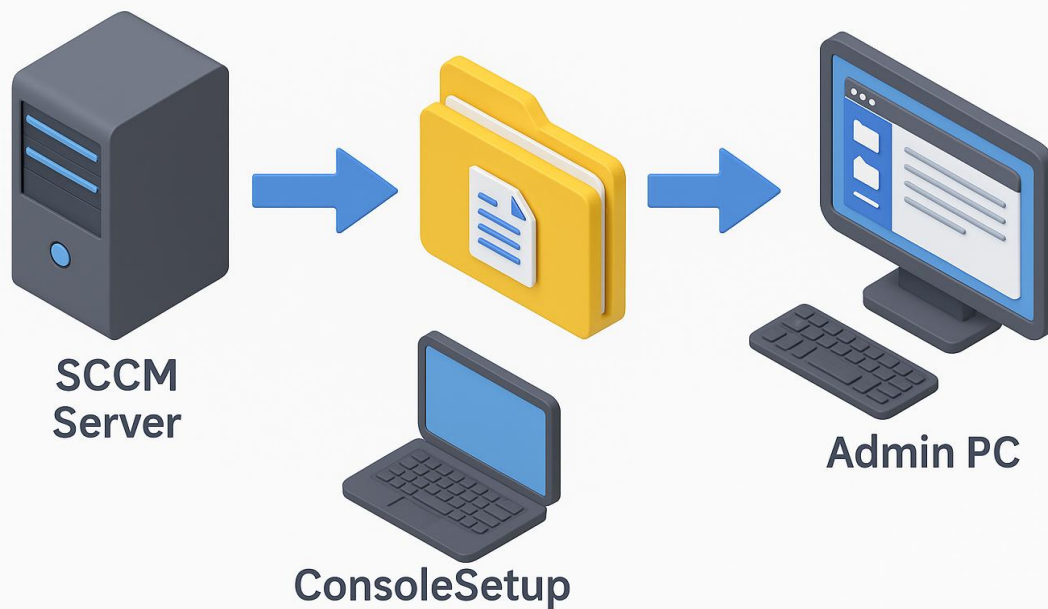
◆ Example Scenario

- A Helpdesk engineer wants to monitor deployments.
- Instead of connecting to the SCCM server, he installs **SCCM Console** on his own PC.
- Logs in → sees only the features assigned to his role (thanks to RBA).

☑ Summary

- SCCM Console can be installed on remote admin PCs.
- Needs **ConsoleSetup.exe** + proper permissions.
- Improves efficiency, security, and teamwork.

Accessing SCCM Console From A Different Computer



4. Resource Explorer in SCCM

◊ What is Resource Explorer?

- **Resource Explorer** is a built-in SCCM tool that allows IT admins to view **detailed information about devices (clients)** in the SCCM console.
- It provides real-time details on **hardware, software, users, and policies** for a selected device.

👉 Example:

- You want to check a specific PC's **installed apps, OS version, RAM, or last logon user** → Use **Resource Explorer**.

◆ What Information Can You See?

- 1. Hardware Information**
 - Processor, RAM, Hard Disk, BIOS, Serial Number
 - Network details (MAC, IP Address)
- 2. Software Information**
 - Installed applications
 - Software updates installed
 - Pending deployments
- 3. User Information**
 - Last logged-on user
 - Domain information
- 4. Policies & Inventory**

- Compliance settings applied
- Inventory data (Hardware & Software Inventory)

◆ Steps to Access Resource Explorer

1. Open **SCCM Console** → **Assets and Compliance** → **Devices**.
2. Select a **device** (PC or laptop).
3. Right-click → **Start** → **Resource Explorer**.
4. Browse categories:
 - Hardware → Summary, CPU, Memory, Disk, Network
 - Software → Installed Applications, Executables
 - Users → Logon history

◆ Example Use Cases

- **Troubleshooting:** Check if a device has enough RAM for a new app.
- **Inventory Reporting:** Verify if Office 365 is installed on a device.
- **User Tracking:** See who last logged into a machine.

☑ Summary

- **Resource Explorer = Device-level deep dive tool.**
- Shows **hardware, software, users, and compliance data.**
- Helps with **inventory, troubleshooting, and audits**

Resource Explorer



5.Windows 10 Servicing Model in SCCM

◆ What is the Windows 10 Servicing Model?

- Microsoft introduced a new **servicing model** with Windows 10 → Instead of big OS upgrades every few years, Windows 10 gets **regular feature updates + monthly quality updates**.
- SCCM helps organizations **manage, deploy, and monitor** these updates.

◆ Types of Updates in Servicing Model

1. **Feature Updates**

Updates

 - Released twice a year (21H2, 22H2, etc.)
 - Add new features and improvements.
 - Similar to a mini-upgrade (like moving from Windows 10 21H1 → 21H2).

2. Quality

Updates

- Monthly security & bug fix updates (Patch Tuesday).
- Cumulative updates (include all fixes since last update).

◆ Servicing Channels in Windows 10

1. **General Availability Channel (GAC)** → Default channel for most users (stable).
2. **Long-Term Servicing Channel (LTSC)** → For critical systems (no frequent feature updates, only security fixes).

Example:

- Banking ATMs or medical devices → LTSC
- Office laptops → GAC

◆ Managing Servicing with SCCM

- SCCM provides a **Windows 10 Servicing Dashboard**:
 - Shows which version of Windows 10 devices are running.
 - Highlights devices that are **nearing end of support**.
- IT admins can:
 - Create **Servicing Plans** → Automate deployment of feature updates.
 - Deploy **Quality Updates** via Software Updates or ADR (Automatic Deployment Rules).

◆ Steps to Configure Windows 10 Servicing in SCCM

1. Open **SCCM Console** → **Software Library** → **Windows 10 Servicing**.

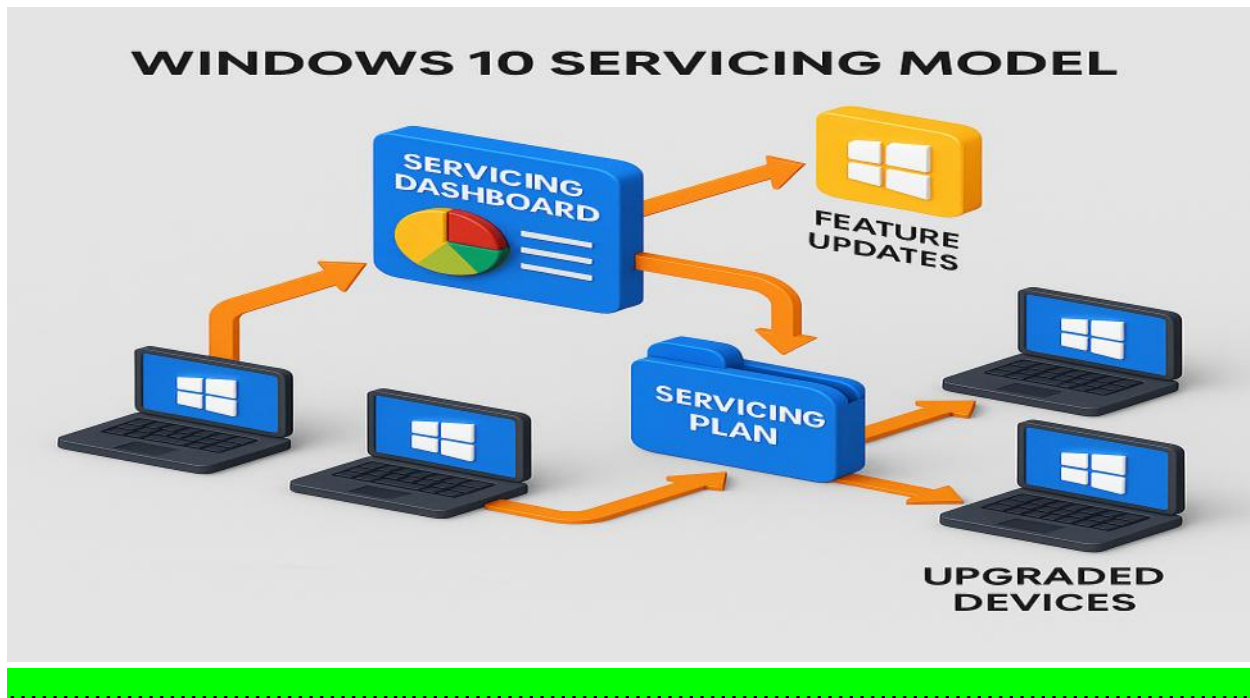
2. View **Servicing Dashboard** (current versions + end-of-life dates).
3. Create a **Servicing Plan**:
 - Define target collection (e.g., All Laptops).
 - Choose which **feature update** to deploy.
 - Set deadlines and schedules.
4. Deploy → Clients download feature update and upgrade in place.

◇ Example Scenario

- Company wants all PCs to move from **Windows 10 21H1** → **22H2**.
- Admin creates a **Servicing Plan** in SCCM.
- Devices automatically receive the upgrade in the next maintenance window.

☑ Summary

- Windows 10 Servicing Model = Continuous updates (Feature + Quality).
- SCCM Servicing Dashboard helps track versions and end-of-support.
- Servicing Plans automate feature update deployments.
- Ensures all PCs remain **secure, compliant, and supported**.



6. Office 365 Client Update Deployment in SCCM

◆ Why Office 365 Updates are Important?

- Microsoft releases **monthly updates** for Microsoft 365 Apps (Word, Excel, Outlook, Teams integration, etc.).
- Updates include **new features, bug fixes, and security patches**.
- SCCM helps IT admins manage these updates centrally to ensure all devices stay secure and up to date.

◆ Update Channels for Office 365 (Microsoft 365 Apps)

1. **Current Channel** → Latest features, released monthly (for general users).
2. **Monthly Enterprise Channel** → Stable monthly updates, tested version.
3. **Semi-Annual Enterprise Channel** → Updates twice a year (for enterprises needing more testing).

👉 Example:

- Developers → Current Channel
- Finance Department → Semi-Annual Enterprise Channel

◆ How SCCM Manages Office 365 Updates

- SCCM integrates with **Microsoft Update Catalog** to download Office 365 updates.
- Updates are deployed like normal **Software Updates**.
- Can be automated using **Automatic Deployment Rules (ADR)**.

◆ Steps to Deploy Office 365 Updates in SCCM

1. Enable Office 365 Updates in SUP (Software Update Point)

- Go to **Administration** → **Site Configuration** → **Sites** → **Configure SUP**.
- Enable **Office 365 Client Updates** under **Products**.

2. Synchronize Software Updates

- Go to **Software Library** → **All Software Updates**.
- Run **sync** to pull Office 365 updates from Microsoft.

3. Create a Deployment Package

- Select required Office 365 update.
- Download content and distribute to **Distribution Points (DPs)**.

4. Deploy Updates

- Deploy to a **device collection** (e.g., “All Office PCs”).
- Choose installation deadline & user experience options.

5. Monitor

Deployment

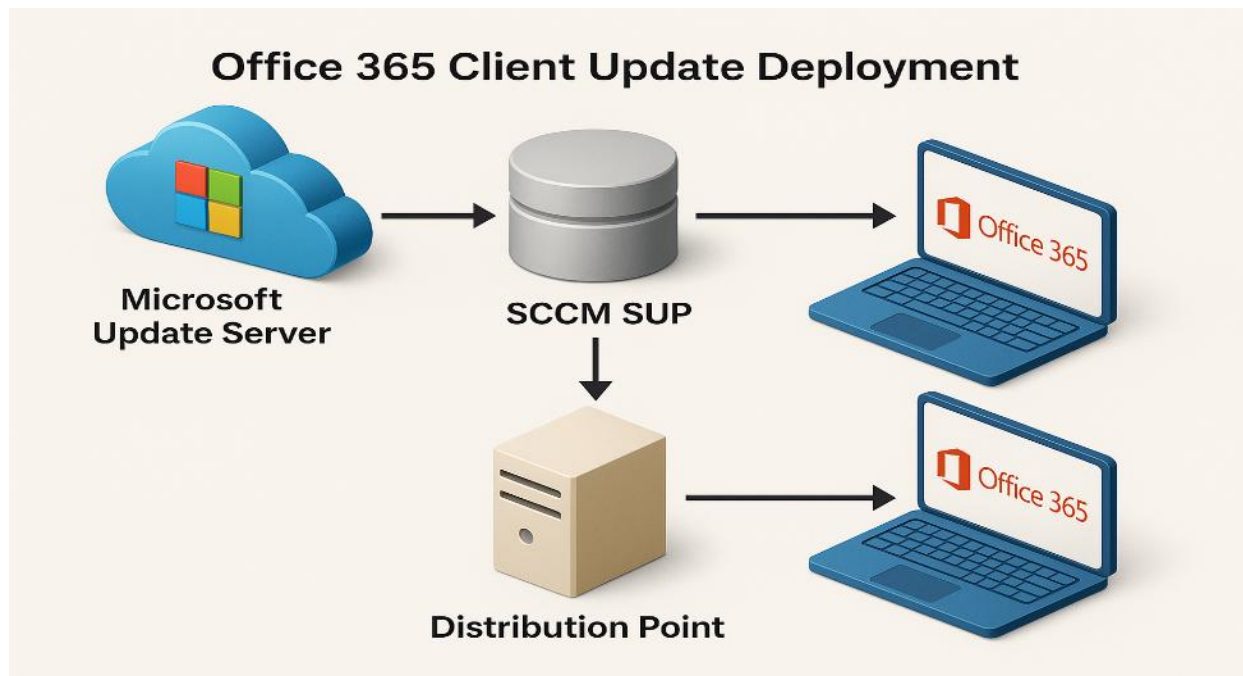
- Check **Monitoring** → **Deployments** for success/failure.
- Review logs on client:
 - `UpdatesHandler.log`
 - `UpdatesDeployment.log`

◇ Example Scenario

- Admin creates a deployment for **Office 365 July Security Update**.
- Deploys to “HR and Finance PCs” collection.
- SCCM pushes the update → Clients download from DP → Office apps updated silently.

☑ Summary

- Office 365 apps need **monthly updates** for security & new features.
- SCCM can deploy updates via **SUP** + **ADR**.
- Different channels (Current, Monthly Enterprise, Semi-Annual) for different needs.
- Easy to monitor via reports & logs.



7. Upgrading SCCM from Old Version to New Version

◆ Why Upgrade SCCM?

- Microsoft regularly releases **new versions** of SCCM (Configuration Manager).
- Upgrading ensures:
 - Support for **latest Windows versions** (Windows 11, Server 2022).
 - Support for **new features** (modern management, cloud integration, analytics).
 - Latest **security fixes** and **bug patches**.

◆ Upgrade Types

1. In-Place

Upgrade

- Upgrade SCCM directly on the existing server.
- Keeps existing roles, DB, and settings.

- Most commonly used.

2. Side-by-Side Migration

- Install new SCCM version on a new server → Migrate settings & clients.
- Used for **big version jumps** or hardware replacement.

◆ Prerequisites Before Upgrade

- ✓ Backup SCCM database & site server.
- ✓ Apply all critical Windows Updates on the server.
- ✓ Check **disk space** (need at least 15–20 GB free).
- ✓ Update ADK (Windows Assessment and Deployment Kit) to latest.
- ✓ Upgrade SQL Server if required (SCCM version-specific).
- ✓ Verify replication health (Primary ↔ Secondary sites).

◆ Steps for In-Place Upgrade

1. **Download SCCM Latest Version** from Microsoft Volume Licensing or MECM site.
2. Run **Splash.hta** → Launch **SCCM Setup Wizard**.
3. Choose **Upgrade this Configuration Manager site**.
4. SCCM runs a **Prerequisite Check** → Fix issues if any.
5. Continue upgrade → Core components, DB, and console upgraded.
6. Post-upgrade tasks:
 - Upgrade **Boot Images**.
 - Update **Client Agents** (push new version to all clients).
 - Validate **custom reports, task sequences, boundaries**.

◆ Monitoring the Upgrade

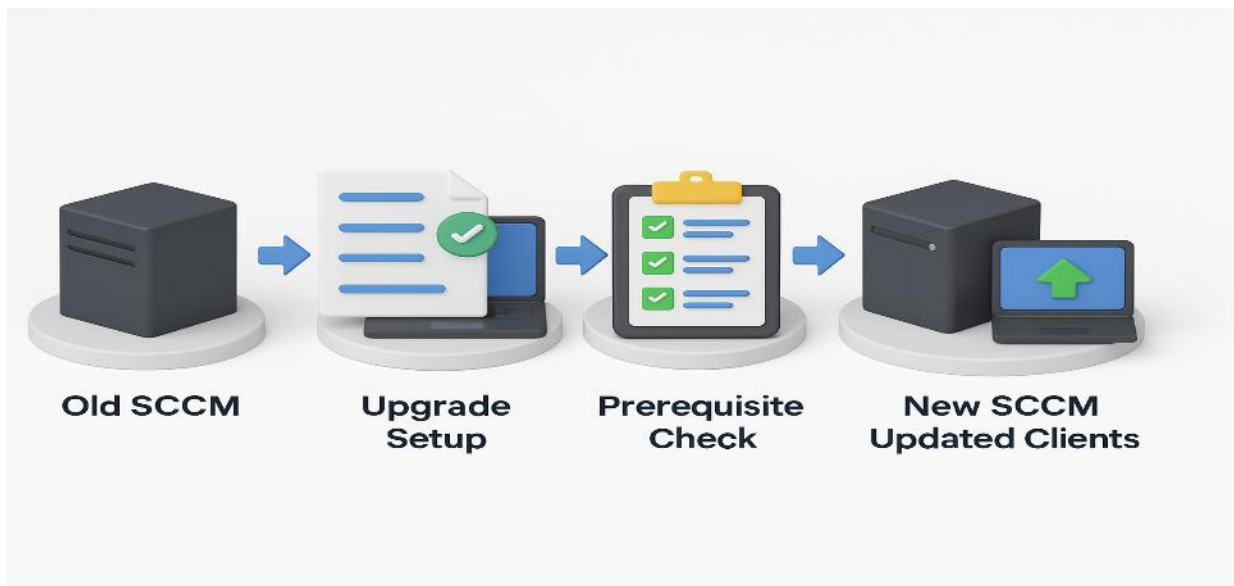
- Logs → to monitor:
 - `CMUpdate.log` → Upgrade process details.
 - `ConfigMgrSetup.log` → Setup progress.
 - `hman.log` → Site component upgrade.
- Console → **Monitoring** → **Updates and Servicing Status** → shows progress.

◇ Example Scenario

- SCCM 2107 → Upgrade to SCCM 2207.
- Admin runs upgrade → 2 hours downtime → SCCM upgraded.
- Clients automatically get new agent version.

☑ Summary

- Upgrade keeps SCCM environment secure & feature-rich.
- In-place upgrade is most common.
- Always **backup, check prerequisites, and test in lab** before production upgrade.
- After upgrade → update boot images, client agents, and validate site health.



Module 16: Placement Guide

1. Tips to Clear an Interview (SCCM / IT Admin Roles)

◆ 1. Build Strong Fundamentals

- **Revise** **core** **SCCM** **concepts:**
 - Site hierarchy (Primary, Secondary)
 - Boundaries & Boundary Groups
 - Application vs Package model
 - Software Updates & ADR
 - OSD (Task Sequences)
 - Logs & Troubleshooting
- Interviewers often test **practical knowledge**, not just theory.

◆ 2. Hands-On Practice

- Set up a **lab environment** (VMs with SCCM + SQL + Windows 10/11 clients).
- Practice:
 - Deploying an app (like Chrome/Office)
 - Creating a servicing plan
 - OSD task sequence
 - Remote control session
- Real experience = More confidence.

◆ 3. Be Clear with Logs & Troubleshooting

- Many interviewers ask: “**How do you troubleshoot failed deployments?**”
- Remember key logs:
 - `AppEnforce.log` → App installation
 - `CAS.log` → Content download
 - `UpdatesDeployment.log` → Updates
 - `smsts.log` → OSD issues

👉 If you can name logs confidently → it shows **practical experience**.

◆ 4. Focus on Communication Skills

- Explain concepts **clearly** and **simply**.
- Use **real-world examples** (like branch office scenario for Secondary Site).
- If you don't know an answer → say, “*I'm not sure, but I would troubleshoot by checking logs and monitoring.*” (Shows problem-solving mindset).

◆ 5. Prepare for Common Interview Questions

Some frequent SCCM interview questions:

1. What's the difference between Application Model and Package Model?
2. How do you troubleshoot if an app didn't install?
3. What are Boundaries and Boundary Groups?
4. How do you deploy Windows Updates?
5. Explain the steps in Operating System Deployment.
6. Which logs do you check for software installation failure?
7. What is Role-Based Administration in SCCM?
8. How do you monitor SCCM Site Health?

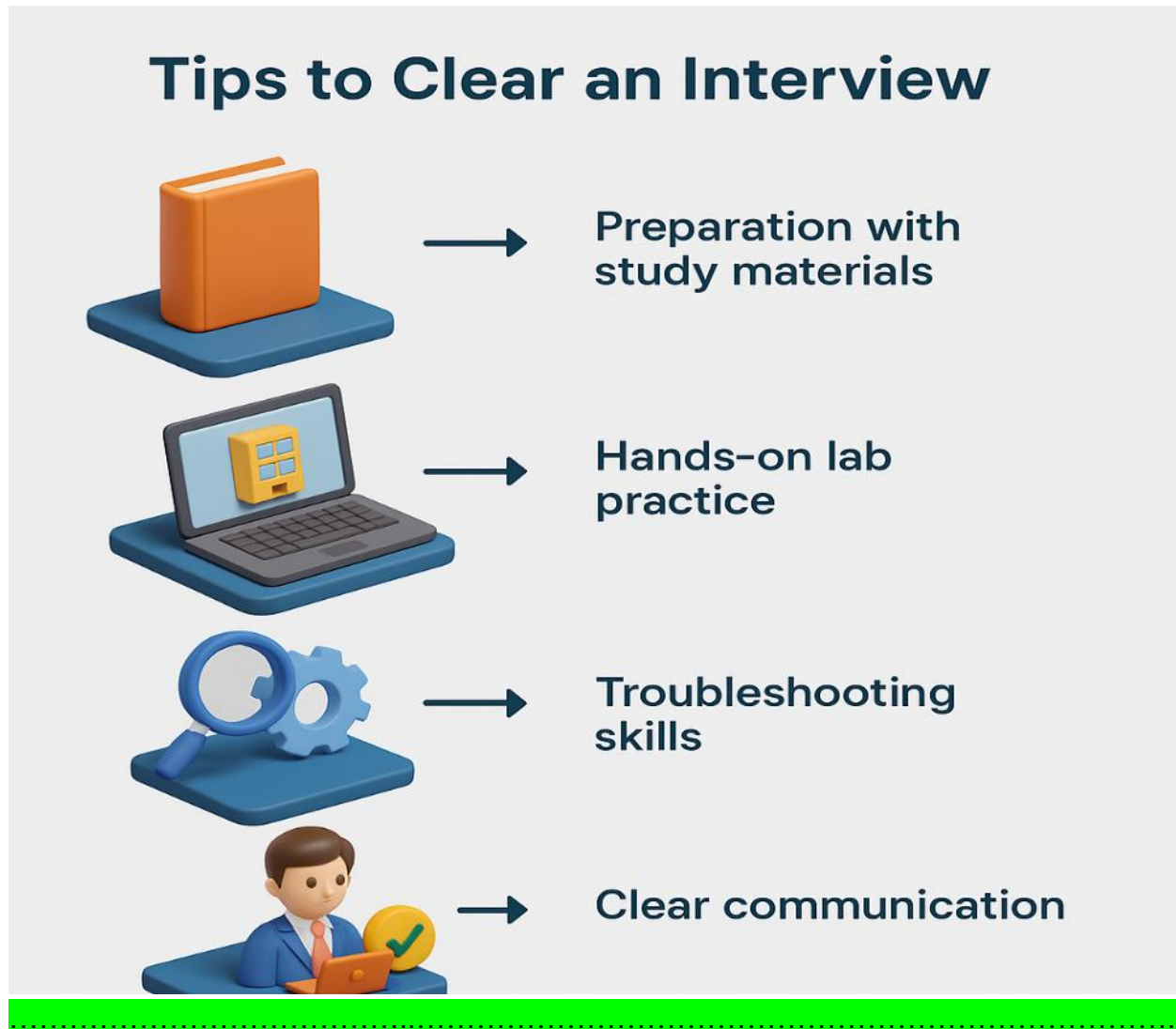
◆ 6. Be Confident in Resume + Projects

- Mention **hands-on projects** (even lab work counts).
- Example: *"Configured SCCM to deploy Office 365 updates to 100+ devices in lab environment."*
- This makes you stand out vs only theory candidates.

☒ Summary:

- Revise fundamentals
- Practice in lab
- Be strong in logs & troubleshooting
- Communicate clearly with examples

- Prepare common SCCM Q&A
- Highlight hands-on projects in resume



2. Common SCCM Interview Questions and Answers

◇ 1. What is SCCM?

Answer:

SCCM (System Center Configuration Manager) is a Microsoft product used for **managing devices** (PCs, laptops, servers, mobiles). It helps in:

- Deploying applications

- Pushing Windows/Office updates
- Operating System Deployment (OSD)
- Managing compliance and security

◇ 2. Difference between Application Model and Package Model?

Answer:

- **Application Model (New)** → Used for modern deployments, supports **detection methods, requirements, user-based targeting.**
- **Package Model (Legacy)** → Older method, used for simple software deployments without detection methods.

👉 Example:

- Use **Application Model** for deploying Office 365.
- Use **Package Model** for scripts or simple EXE installs.

◇ 3. What are Boundaries and Boundary Groups?

Answer:

- **Boundary** → Defines a network location (IP Subnet, IP Range, AD Site).
- **Boundary Group** → Logical collection of boundaries, used to assign **Distribution Points (DPs)** and **site assignment.**

◇ 4. What are the different Site Roles in SCCM?

Answer:

- **Management Point (MP):** Client communication.

- **Distribution Point (DP):** Stores and distributes content.
- **Software Update Point (SUP):** Syncs Windows/Office updates.
- **Reporting Services Point (RSP):** For SCCM reports.

◇ 5. Which logs do you check if an application fails to install?

Answer:

On the client machine:

- `AppDiscovery.log` → Detects if app already installed.
- `CAS.log` → Content download.
- `AppEnforce.log` → App install execution.

On the server:

- `distmgr.log` → Content distribution to DP.

◇ 6. How do you deploy Windows Updates in SCCM?

Answer:

1. Configure **Software Update Point (SUP)**.
2. Sync updates from Microsoft.
3. Create **Update Groups**.
4. Create **Deployment Package** and distribute to DPs.
5. Deploy updates to **device collections**.
6. Monitor deployment status.

◇ 7. Explain Operating System Deployment (OSD).

Answer:

1. Capture **Reference Image** (golden OS).
2. Create **Task Sequence** in SCCM.
3. Deploy Task Sequence to collection.
4. Clients boot via PXE or media → Task Sequence installs OS, drivers, apps, policies.

◇ 8. What is Role-Based Administration (RBA)?

Answer:

RBA controls **who can do what** in SCCM.

- **Security Role** → Defines what actions user can do.
- **Security Scope** → Defines which objects they can manage.
- **Collection** → Defines target devices/users.

◇ 9. What is Software Metering in SCCM?

Answer:

- Feature that tracks **application usage** (how many times, how long).
- Helps reduce licensing costs.
👉 Example: Track if MS Project is really being used or not.

◇ 10. How do you monitor SCCM site health?

Answer:

- **Monitoring Workspace** → Check Site & Component Status.
- Use **Reports** for system role availability.
- Check key logs like `hman.log`, `mpcontrol.log`, `distmgr.log`.

☒ **Summary for Interviews:**

- Revise **basics** (what is **SCCM**, **site** **roles**)
- Be strong in **Boundaries**, **Applications vs Packages**, **OSD**, **Updates**
- Always mention **logs** when troubleshooting → shows hands-on experience.

Common SCCM Interview Questions and Answers



What is SCCM?



Difference between Application Model and Package Model?



What are Boundaries and Boundary Groups?



What are the different Site Roles in SCCM?



Which logs do you check if an application fails to install?



How do you deploy Windows Updates in SCCM?



Explain Operating System Deployment (OSD).



What is Role-Based Administration (RBA)?

3. Microsoft SCCM 2016 Interview Questions and Answers

◆ 1. What is SCCM 2016 used for?

Answer:

SCCM 2016 (System Center Configuration Manager) is used to centrally manage devices:

- Deploying applications and software updates
- Operating system deployment (OSD)
- Patch management
- Compliance and inventory
- Endpoint protection

◆ 2. What's the difference between SCCM 2012 and SCCM 2016?

Answer:

- SCCM 2016 introduced the “**Current Branch**” model (frequent updates).
- Better integration with **Windows 10 Servicing**.
- Enhanced support for **Office 365 updates**.
- Improved cloud integration with **Intune (MDM)**.
- More role-based administration features.

◆ 3. What are Boundaries and Boundary Groups in SCCM 2016?

Answer:

- **Boundary** → Defines a network location (IP subnet, AD site, IP range).

- **Boundary Group** → Logical grouping of boundaries → used to assign Distribution Points (DPs) and site assignment.

◆ 4. Explain Primary Site vs Secondary Site in SCCM 2016.

Answer:

- **Primary Site** → Full site that manages clients, stores database, and supports up to 100,000 clients.
- **Secondary Site** → Dependent site, used for branch offices to reduce WAN traffic. Uses SQL Express.

◆ 5. What are the different site system roles in SCCM 2016?

Answer:

- **Management Point (MP):** Client communication.
- **Distribution Point (DP):** Stores apps, OS, updates.
- **Software Update Point (SUP):** Syncs Microsoft updates.
- **Reporting Services Point (RSP):** For SSRS reports.
- **Fallback Status Point (FSP):** Helps troubleshoot client issues.

◆ 6. Which logs are important in SCCM 2016 troubleshooting?

Answer:

- **Server-side:**
 - `hman.log` (Hierarchy Manager)
 - `distmgr.log` (Distribution Manager)
 - `mpcontrol.log` (MP health)

- `wsyncmgr.log` (WSUS sync)

- **Client-side:**

- `AppDiscovery.log`, `AppEnforce.log` (Application deployment)
- `CAS.log` (Content download)
- `smsts.log` (OSD)
- `UpdatesDeployment.log` (Patch deployment)

◆ **7. How do you deploy Office 365 updates in SCCM 2016?**

Answer:

1. Enable **Office 365 Client Updates** in SUP settings.
2. Sync software updates.
3. Create a deployment package.
4. Distribute to DPs.
5. Deploy to target device collections.

◆ **8. What is Role-Based Administration (RBA)?**

Answer:

- **Security Roles** → Define what a user can do.
- **Security Scopes** → Define which objects a user can manage.
- **Collections** → Define which devices/users a user can target.

👉 Example: A Helpdesk Engineer can only use **Remote Control**, but cannot deploy apps.

◆ **9. What is the Windows 10 Servicing Dashboard in SCCM 2016?**

Answer:

- A feature that shows the **Windows 10 versions** running in the environment.
- Highlights devices that are nearing **end of support**.
- Allows admins to create **Servicing Plans** to automatically deploy feature updates.

◆ **10. What's the process to upgrade SCCM from 2012 to 2016?**

Answer:

1. Backup SCCM DB & server.
2. Upgrade SQL Server to supported version.
3. Run **SCCM 2016 setup** → Choose **Upgrade this site**.
4. Perform **prerequisite check**.
5. Run upgrade.
6. Post-upgrade tasks: Update boot images, client agents, validate components.

☒ **Summary for SCCM 2016 Interviews:**

- Be confident with **Boundaries, Applications vs Packages, OSD, Updates**.
- Always highlight **logs** when explaining troubleshooting.
- Mention **new features of SCCM 2016** (Windows 10 servicing, Office 365 updates, Current Branch model).



4. Resume Building Guide (for SCCM / IT Admin / Data Analyst Roles)

◆ 1. Keep It Clean and Simple

- Use a **professional format** (no fancy colors, only black/blue text).
- Stick to **1–2 pages max.**
- Use clear headings: *Summary, Skills, Experience, Projects, Education.*

◆ 2. Write a Strong Summary (3–4 lines)

Example:

"SCCM and IT Infrastructure Administrator with hands-on experience in managing application deployment, patch management, OSD, and compliance settings. Skilled in troubleshooting, log analysis, and implementing efficient IT solutions. Strong interest in Microsoft technologies and enterprise IT operations."

◆ 3. Highlight Core Skills

👉 For SCCM / IT Roles, include:

- SCCM (Applications, Packages, OSD, SUP, DP, MP, RBA)
- Windows 10/11 Servicing & Office 365 Updates
- SQL Server (basic queries, DB backup/restore)
- Active Directory & Group Policy
- PowerShell scripting (basic automation)
- Troubleshooting with logs

👉 For Data Analyst roles (since you also aim for that):

- Excel, Power BI, Tableau
- SQL, Python, Pandas, NumPy
- Data Cleaning & Visualization
- Dashboards & KPI Reporting

◆ 4. Show Hands-On Experience (Even from Lab)

Example projects to mention:

- *"Configured SCCM to deploy Office 365 updates to 100+ devices in lab."*
- *"Created Windows 10 Servicing Plan to upgrade clients from 21H1 to 22H2."*

- “Implemented compliance policies for password & encryption via SCCM.”
- “Built Power BI dashboard for sales analysis using SQL data.”

Even lab work counts as **practical experience** for interviews.

◆ 5. Mention Certifications (if any)

- Microsoft Certified: Endpoint Administrator (MD-102)
- ITIL Foundation (if applicable)
- SQL / Data Analytics certificates (Coursera, Microsoft Learn, etc.)

◆ 6. Education Section

- Keep it simple: *BCA Graduate (Year, University).*

◆ 7. Extra Tips

- ✓ Use **bullet points** (not long paragraphs).
- ✓ Start each point with **Action Verb**: Configured, Deployed, Implemented, Managed, Designed.
- ✓ Avoid spelling/grammar mistakes.
- ✓ Tailor resume for **each job role** (SCCM Admin vs Data Analyst).

☒ Example Resume Structure

1. **Summary**
Short intro (3–4 lines).

2. **Skills**
List SCCM + IT + Data skills.

3. Experience / Projects

- Bullet points (Action Verb + Task + Result).

4.
Degree + University.

Education

5. **Certifications**
Relevant IT/Analytics certificates.

(Optional)



5. How to Attempt an Interview (SCCM / IT / Data Analyst Roles)

◆ 1. Before the Interview

- ✓ Revise **core concepts** (SCCM basics, site roles, boundaries, OSD, updates, logs).
- ✓ Practice **mock questions** aloud (confidence aata hai).
- ✓ Keep your **resume printed + soft copy** ready.
- ✓ Dress professionally (formal shirt, neat appearance).

◆ 2. During the Interview

- First Impression Matters**
 - Greet politely (*"Good Morning Sir/Madam"*).
 - Sit confidently (not stiff, not casual).
- Answering Questions**
 - Listen carefully before answering.
 - Use **STAR method** (Situation → Task → Action → Result).
 - If unsure, say: *"I'm not sure, but I will check logs / test in lab."*
- Show Practical Knowledge**
 - Use **real examples** (e.g., *"I checked AppEnforce.log to troubleshoot failed Chrome installation"*).
- Communication Style**
 - Speak slowly, clearly, simple English.
 - Avoid yes/no answers → add short explanation.

◆ 3. Common Mistakes to Avoid

- ✗ Don't memorize answers blindly.
- ✗ Don't argue if you don't know something.
- ✗ Don't say "I don't know anything about that." Instead say, *"I've not worked directly, but I understand the concept."*

◆ 4. After the Interview

- Thank the panel (*"Thank you for the opportunity"*).
- If asked, be ready with **expected salary** and **notice period**.
- Stay confident even if you don't know some answers.

✓ Golden Tip

💡 Interviewers don't just test knowledge, they test problem-solving + confidence. Even if you know **70%**, but show **confidence and logical approach**, you can clear it.

